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Excel -
Advanced Data Analysis

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Agenda

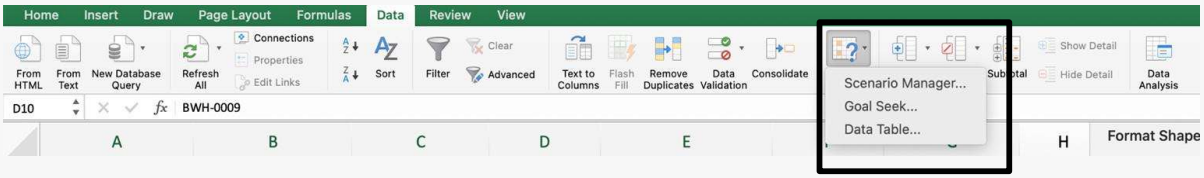
- What if Analysis
- Data Table
- Scenario Manager
- Pivot Table
- Goal Seek

DQLab

What if Analysis Excel

Fitur ini adalah kumpulan beberapa perintah Microsoft Excel yang umum diterapkan di lingkungan bisnis. biasanya digunakan oleh user dengan level managerial keatas. Sesuai dengan namanya yaitu **What If Analysis, atau disebut juga Analisa Bagaimana Jika,** perintah ini bisa kita gunakan untuk membuat suatu analisa atau prediksi tentang berbagai kemungkinan yang mungkin terjadi dalam tabel perhitungan.

Didalam fitur **What If Analysis**, terdapat beberapa **sub fitur** seperti **Scenario Manager, Goal Seek, Data Table, Solver, Pivot Table**



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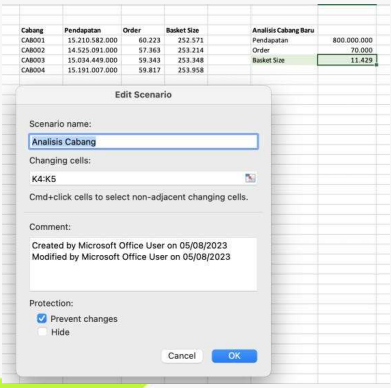
Scenario Manager

Scenario Manager adalah salah satu fitur data analysis yang dapat anda gunakan untuk melihat suatu hasil atau output dari beberapa input yang berbeda. Perintah ini digunakan untuk analisa bisnis atau analisa suatu project dengan cara melihat beragam kemungkinan perhitungan yang mungkin terjadi atau sebagai perbandingan suatu nilai output dari berbagai nilai inputan.



Cabang	Pendapatan	Order	Basket Size
CAB001	15.210.582.000	60.223	252.571
CAB002	14.525.091.000	57.363	253.214
CAB003	15.034.449.000	59.343	253.348
CAB004	15.191.007.000	59.817	253.958

Analisis Cabang Baru	
Pendapatan	800.000.000
Order	70.000
Basket Size	11.429

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Goal Seek

Goal Seek merupakan salah satu cara penerapan perhitungan yang disediakan oleh Excel dalam membantu perhitungan secara efektif dan efisien. Fasilitas ini fleksibel untuk berbagai kasus yang ada, khususnya yang membutuhkan perhitungan akurat dalam perencanaan guna pengambilan keputusan.

Cabang	Pendapatan	Order	Basket Size
CAB001	15.210.582.000	60.223	252.571
CAB002	14.525.091.000	57.363	253.214
CAB003	15.034.449.000	59.343	253.348
CAB004	15.191.007.000	59.817	253.958
CAB005	10.038.871.000		
Average	**14.000.000.000**	**59.187**	**253.273**

Cabang	Pendapatan	Order	Basket Size
CAB001	15.210.582.000	60.223	252.571
CAB002	14.525.091.000	57.363	253.214
CAB003	15.034.449.000	59.343	253.348
CAB004	15.191.007.000	59.817	253.958
CAB005	10.038.871.000	48.254	246.909
Average	**14.000.000.000**	**57.000**	**252.000**

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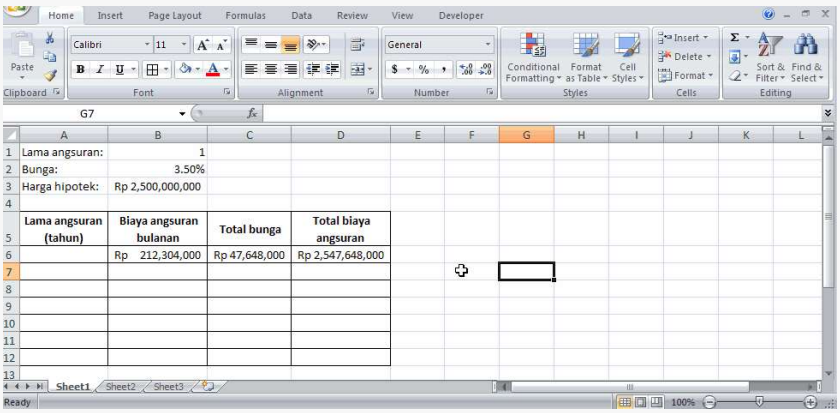
Data Table – One Variable and Two Variable



Data Table adalah salah satu perintah dari fitur What-If Analysis selain Scenario Manager yang sangat bermanfaat untuk dapat menampilkan hasil dari suatu rumus berdasarkan nilai-nilai input alternatif yang ditetapkan.

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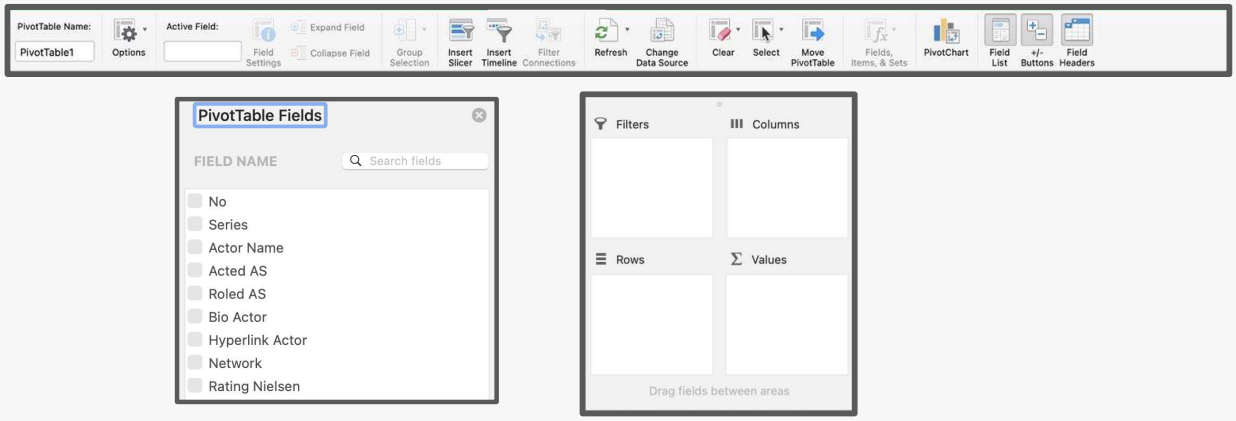
Pivot Table- Multi Variable



PivotTable adalah salah satu tools excel yang digunakan untuk menghitung, meringkas, dan menganalisis data yang juga dapat digunakan untuk melihat perbandingan, pola, dan tren dalam data.

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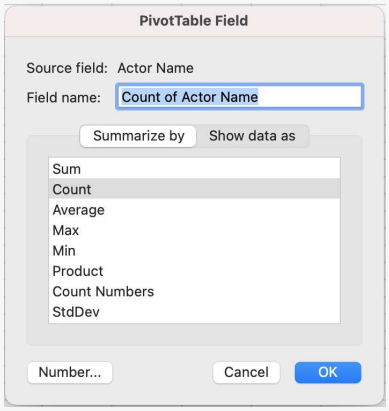
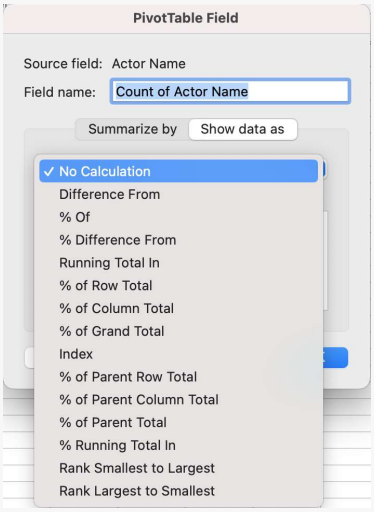
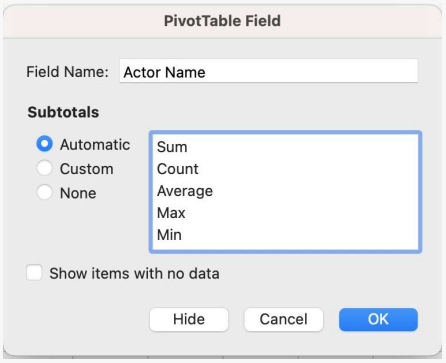
Pivot Table- Multi Variable







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Hands-on
Sesi 12

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Goal Seek

Toko **DQLab Fashion** menargetkan laba (**income**) sebesar **Rp150.000.000** dari penjualan satu produk. Harga jual produk adalah **Rp21.000**, dan **biaya pokok penjualan (COGS)** adalah **Rp15.000** per unit.

Berapakah jumlah produk yang harus terjual untuk mencapai target income tersebut?

	A	B	C	D	E	F
1		**Goal Seek**				
2						
3		**Jumlah Produk Terjual**	16,667			
4		**Harga per Produk**	21,000			
5		**Omset**				
6						
7		**COGS per produk**	15,000			
8		**Total Cost**				
9						
10		**Income**			**Target Income**	150,000,000
11						
12		**Solver (Advanced Goal Seek)**				
13						
14		**Jumlah Produk Terjual**	12,500			
15		**Harga per Produk**	22,000			
16		**Omset**				
17						
18		**COGS per produk**	14,000			
19		**Total Cost**				
20						
21		**Income**				
22						

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Goal Seek

Instruksi Goal Seek

1. Masukkan rumus:
   - **C5** → `=C3*C4` *(Omset = Jumlah × Harga)*
   - **C8** → `=C3*C7` *(Total Cost = Jumlah × COGS)*
   - **C10** → `=C5-C8` *(Income = Omset – Cost)*
2. Klik sel **C10** (Income)
3. Buka tab **Data** → klik **What-If Analysis** → pilih **Goal Seek**
4. Isi dialog Goal Seek:
   - **Set Cell**: `C10`
   - **To Value**: `150000000`
   - **By Changing Cell**: `C3`
5. Klik OK. Excel akan menghitung dan menampilkan jumlah produk terjual yang diperlukan.

	A	B	C	D	E	F
1		**Goal Seek**				
2						
3		**Jumlah Produk Terjual**	16,667			
4		**Harga per Produk**	21,000			
5		**Omset**	350,000,000	=C3*C4		
6						
7		**COGS per produk**	15,000			
8		**Total Cost**	250,000,000	=C7*C3		
9						
10		**Income**	**100,000,000**	=C5-C8	**Target Income**	150,000,000
11						

	A	B	C	D
1		**Goal Seek**		
2				
3		**Jumlah Produk Terjual**	25,000	
4		**Harga per Produk**	21,000	
5		**Omset**	525,000,000	=C3*C4
6				
7		**COGS per produk**	15,000	
8		**Total Cost**	375,000,000	=C7*C3
9				
10		**Income**	**150,000,000**	=C5-C8
11				

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Solver

Toko DQLab Fashion menargetkan **income sebesar Rp150.000.000** dari penjualan satu produk.

- **Harga per Produk** dan **Jumlah Produk Terjual** masih dapat disesuaikan.
- **COGS per Produk** tetap: **Rp14.000**
  Gunakan **Solver** untuk menghitung kombinasi **harga jual** dan **jumlah produk** yang tepat.

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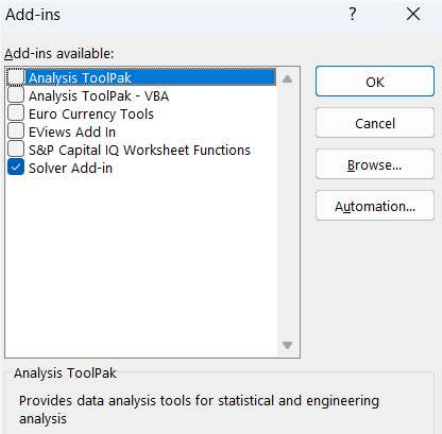
Solver

✅ Langkah-langkah Mengaktifkan Solver di Excel:

1. Klik **File** (pojok kiri atas)
2. Pilih **Options**
3. Di jendela **Excel Options**, pilih **Add-ins**
4. Di bagian bawah, pastikan **Manage: Excel Add-ins** → klik **Go...**
5. Centang **Solver Add-in**, lalu klik **OK**



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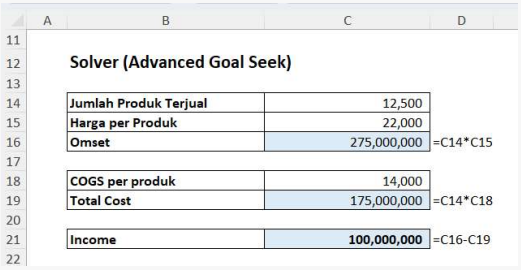
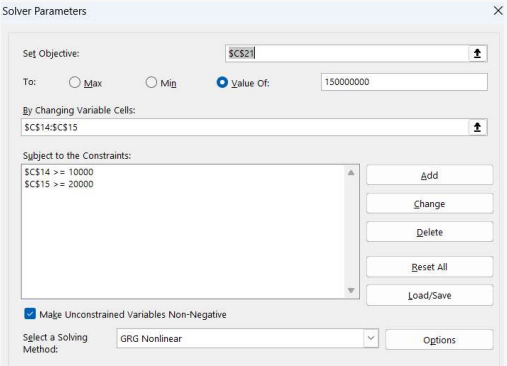
🛠️ Langkah-langkah Menjalankan Solver:

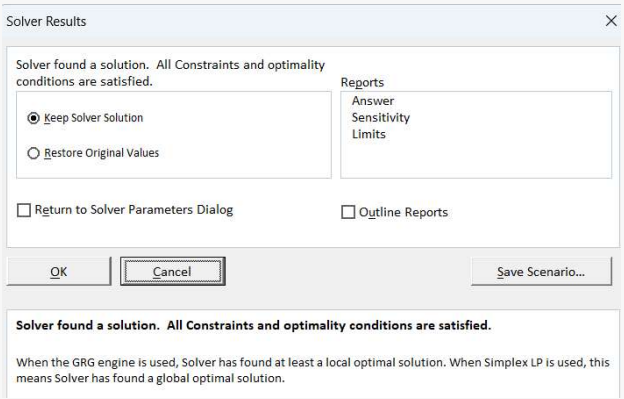
1. **Pastikan rumus berikut dimasukkan:**
   - `C16 = C14 * C15` *(Omset)*
   - `C19 = C14 * C18` *(Total Cost)*
   - `C21 = C16 - C19` *(Income)*
2. Klik menu **Data** → **Solver**
3. Atur parameter Solver:
   - **Set Objective:** `C21`
   - **To Value Of:** `150000000`
   - **By Changing Variable Cells:** `C14, C15`
4. (Opsional) Tambahkan batasan:
   - `C14 >= 10000` *(Jumlah minimum 10.000)*
   - `C15 >= 20000` *(Harga minimum Rp20.000)*
5. Klik **Solve**
6. Setelah berhasil, pilih **Keep Solver Solution** → klik **OK**



Solver (Advanced Goal Seek)

Jumlah Produk Terjual	12,500	
Harga per Produk	22,000	
Omset	275,000,000	=C14*C15
COGS per produk	14,000	
Total Cost	175,000,000	=C14*C18
Income	**100,000,000**	=C16-C19





Solver (Advanced Goal Seek)

Jumlah Produk Terjual	13,196	
Harga per Produk	25,367	
Omset	334,740,144	=C14*C15
COGS per produk	14,000	
Total Cost	184,740,167	=C14*C18
Income	**149,999,977**	=C16-C19

Data Table (Single Variable)

📦 **Konteks:**

Manajer penjualan DQLab Fashion ingin menganalisis **dampak jumlah produk terjual terhadap kinerja keuangan.**

Dengan asumsi harga per produk, variable cost, dan fixed cost sudah ditentukan, manajer ingin mengetahui:

- Omset
- Biaya variabel
- Contribution margin
- Net income
- Total cost

...untuk setiap skenario jumlah produk yang terjual.

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🎯 **Tujuan:**

Hitung dampak perubahan **jumlah produk terjual** terhadap:

- Omset
- Biaya Variabel
- Contribution Margin
- Net Income
- Total Costs

Dengan menggunakan **Data Table satu variabel (Single Variable Data Table)** di kolom F–K.

Assumptions	
Produk Terjual	3,000
Harga per Produk	Rp10,900
Variable Cost per Produk	Rp7,000
Fixed Costs	Rp10,000,000
Jumlah Terjual (Start)	0
Jumlah Terjual (Kelipatan)	250

	D	E	F	G	H	I	J	K
1								
2								
3		Units		Omset	Variable Costs	Contribution Margin	Net Income	Total Costs
4								
5								
6								

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✅ **Langkah 1: Buat rumus utama di blok kiri atas**

Isi sel berikut ini:

	A	B	C	D
1		**Single Variable**		
2				
3		Omset	32,700,000.00	=C12*C13
4		Variable Costs	21,000,000.00	=C12*C14
5		Contribution Margin	11,700,000.00	=C3-C4
6		Fixed Costs	Rp10,000,000	=C15
7		Net Income	1,700,000.00	=C5-C6
8				
9		Total Costs	31,000,000.00	=C4+C6
10				

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✅ Langkah 2: Isi Kolom dengan Units (Jumlah Produk Terjual)

✅ Langkah 3: Hubungkan Sel Output ke Header Baris 2

	D	E	F	G	H	I	J	K	L
1									
2				=C3	=C4	=C5	=C7	=C9	
3		Units		Omset	Variable Costs	Contribution Margin	Net Income	Total Costs	
4		=C12	3,000	32,700,000.00	21,000,000.00	11,700,000.00	1,700,000.00	31,000,000.00	
5		=C17	250						
6		=F5+$C$17	500						
7		=F6+$C$17	750						
8		=F7+$C$17	1000						
9		=F8+$C$17	1250						
10		=F9+$C$17	1500						
11		=F10+$C$1	1750						
12		=F11+$C$1	2000						

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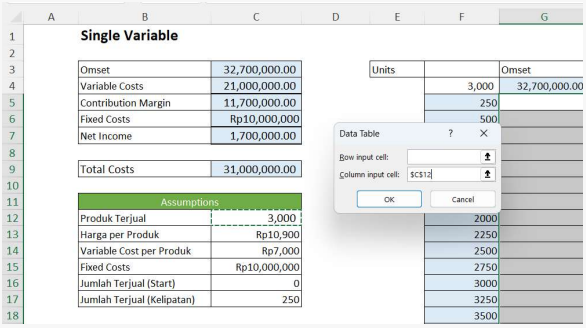
Langkah 3: Buat Data Table

1. **Blok seluruh area dari** F4:K24
2. Klik tab **Data → What-If Analysis → Data Table**
3. Dialog:
   - **Row input cell:** *(kosongkan)*
   - **Column input cell:** `C12` *(karena units mengganti Produk Terjual)*
4. Klik **OK**



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Single Variable

Omset	32,700,000.00
Variable Costs	21,000,000.00
Contribution Margin	11,700,000.00
Fixed Costs	Rp10,000,000
Net Income	1,700,000.00

Total Costs	31,000,000.00

Assumptions	
Produk Terjual	3,000
Harga per Produk	Rp10,900
Variable Cost per Produk	Rp7,000
Fixed Costs	Rp10,000,000
Jumlah Terjual (Start)	0
Jumlah Terjual (Kelipatan)	250

Units		Omset	Variable Costs	Contribution Margin	Net Income	Total Costs
	3,000	32,700,000.00	21,000,000.00	11,700,000.00	1,700,000.00	31,000,000.00
	250	2,725,000	1,750,000	975,000	(9,025,000)	11,750,000
	500	5,450,000	3,500,000	1,950,000	(8,050,000)	13,500,000
	750	8,175,000	5,250,000	2,925,000	(7,075,000)	15,250,000
	1000	10,900,000	7,000,000	3,900,000	(6,100,000)	17,000,000
	1250	13,625,000	8,750,000	4,875,000	(5,125,000)	18,750,000
	1500	16,350,000	10,500,000	5,850,000	(4,150,000)	20,500,000
	1750	19,075,000	12,250,000	6,825,000	(3,175,000)	22,250,000
	2000	21,800,000	14,000,000	7,800,000	(2,200,000)	24,000,000
	2250	24,525,000	15,750,000	8,775,000	(1,225,000)	25,750,000
	2500	27,250,000	17,500,000	9,750,000	(250,000)	27,500,000
	2750	29,975,000	19,250,000	10,725,000	725,000	29,250,000
	3000	32,700,000	21,000,000	11,700,000	1,700,000	31,000,000
	3250	35,425,000	22,750,000	12,675,000	2,675,000	32,750,000
	3500	38,150,000	24,500,000	13,650,000	3,650,000	34,500,000
	3750	40,875,000	26,250,000	14,625,000	4,625,000	36,250,000
	4000	43,600,000	28,000,000	15,600,000	5,600,000	38,000,000
	4250	46,325,000	29,750,000	16,575,000	6,575,000	39,750,000
	4500	49,050,000	31,500,000	17,550,000	7,550,000	41,500,000
	4750	51,775,000	33,250,000	18,525,000	8,525,000	43,250,000
	5000	54,500,000	35,000,000	19,500,000	9,500,000	45,000,000

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Data Table (Double Variable)

📦 **Konteks:**

Manajemen DQLab Fashion ingin mengetahui **dampak kombinasi antara Harga per Produk dan Jumlah Produk Terjual** terhadap **Net Income**.

Parameter biaya tetap dan biaya variabel per unit sudah diketahui. Manajer ingin mensimulasikan beberapa skenario:

- Harga per produk: bervariasi (horizontal)
- Jumlah produk terjual: bervariasi (vertikal)

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🧱 Struktur Rumus Utama (Kolom C)

Double Variable

Omset	32,700,000.00	=C38*C39
Variable Costs	21,000,000.00	=C38*C40
Contribution Margin	11,700,000.00	=C29-C30
Fixed Costs	Rp10,000,000	=C41
Net Income	1,700,000.00	=C31-C32
Total Costs	31,000,000.00	=C30+C32

Assumptions	
Produk Terjual	3,000
Harga per Produk	Rp10,900
Variable Cost per Produk	Rp7,000
Fixed Costs	Rp10,000,000
Jumlah Terjual (Start)	0
Jumlah Terjual (Kelipatan)	250
Harga Jual (Start)	9000
Harga Jual (Kelipatan)	1000

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📈 Langkah 1: Isi Heading (Harga per Produk)

🔁 Langkah 2: Isi Jumlah Produk Terjual di Kolom F

Assumptions	
Produk Terjual	3,000
Harga per Produk	Rp10,900
Variable Cost per Produk	Rp7,000
Fixed Costs	Rp10,000,000
Jumlah Terjual (Start)	0
Jumlah Terjual (Kelipatan)	250
Harga Jual (Start)	9000
Harga Jual (Kelipatan)	1000

		=C44	=G30+$C$45	=H30+$C$45	=I30+$C$45	=J30+$C$45
=C33	1,700,000.00	9,000.00	10,000.00	11,000.00	12,000.00	13,000.00
=C42	0					
=F31+$C$43	250					
=F32+$C$43	500					
=F33+$C$43	750					
=F34+$C$43	1000					
=F35+$C$43	1250					
=F36+$C$43	1500					
=F37+$C$43	1750					
=F38+$C$43	2000					
=F39+$C$43	2250					
=F40+$C$43	2500					
=F41+$C$43	2750					
=F42+$C$43	3000					
=F43+$C$43	3250					
=F44+$C$43	3500					
=F45+$C$43	3750					
=F46+$C$43	4000					
=F47+$C$43	4250					
=F48+$C$43	4500					
=F49+$C$43	4750					

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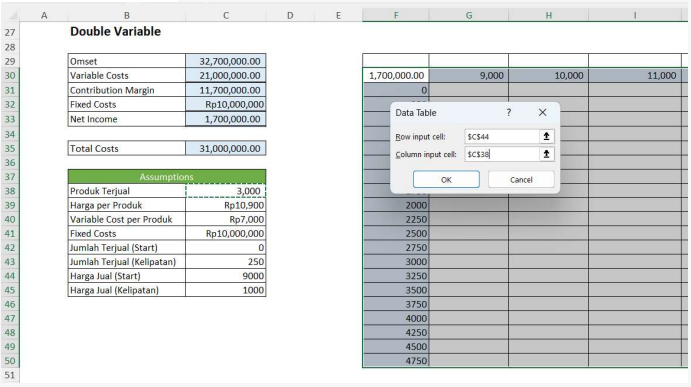
Langkah 4: Buat Data Table

1. **Blok seluruh area dari** F30 **ke kanan dan bawah** (F30:K50)
2. Klik tab **Data → What-If Analysis → Data Table**
3. Di dialog:
   - **Row input cell** C44 *(Harga per Produk)*
   - **Column input cell:** C38 *(Jumlah Produk Terjual)*
4. Klik **OK**



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Double Variable

Omset	32,700,000.00
Variable Costs	21,000,000.00
Contribution Margin	11,700,000.00
Fixed Costs	Rp10,000,000
Net Income	1,700,000.00
Total Costs	31,000,000.00
Assumptions	
Produk Terjual	3,000
Harga per Produk	Rp10,900
Variable Cost per Produk	Rp7,000
Fixed Costs	Rp10,000,000
Jumlah Terjual (Start)	0
Jumlah Terjual (Kelipatan)	250
Harga Jual (Start)	9000
Harga Jual (Kelipatan)	1000

1,700,000.00	9,000.00	10,000.00	11,000.00	12,000.00	13,000.00
0	($10,000,000.00)	($10,000,000.00)	($10,000,000.00)	($10,000,000.00)	($10,000,000.00)
250	($9,500,000.00)	($9,250,000.00)	($9,000,000.00)	($8,750,000.00)	($8,500,000.00)
500	($9,000,000.00)	($8,500,000.00)	($8,000,000.00)	($7,500,000.00)	($7,000,000.00)
750	($8,500,000.00)	($7,750,000.00)	($7,000,000.00)	($6,250,000.00)	($5,500,000.00)
1000	($8,000,000.00)	($7,000,000.00)	($6,000,000.00)	($5,000,000.00)	($4,000,000.00)
1250	($7,500,000.00)	($6,250,000.00)	($5,000,000.00)	($3,750,000.00)	($2,500,000.00)
1500	($7,000,000.00)	($5,500,000.00)	($4,000,000.00)	($2,500,000.00)	($1,000,000.00)
1750	($6,500,000.00)	($4,750,000.00)	($3,000,000.00)	($1,250,000.00)	$500,000.00
2000	($6,000,000.00)	($4,000,000.00)	($2,000,000.00)	$0.00	$2,000,000.00
2250	($5,500,000.00)	($3,250,000.00)	($1,000,000.00)	$1,250,000.00	$3,500,000.00
2500	($5,000,000.00)	($2,500,000.00)	$0.00	$2,500,000.00	$5,000,000.00
2750	($4,500,000.00)	($1,750,000.00)	$1,000,000.00	$3,750,000.00	$6,500,000.00
3000	($4,000,000.00)	($1,000,000.00)	$2,000,000.00	$5,000,000.00	$8,000,000.00
3250	($3,500,000.00)	($250,000.00)	$3,000,000.00	$6,250,000.00	$9,500,000.00
3500	($3,000,000.00)	$500,000.00	$4,000,000.00	$7,500,000.00	$11,000,000.00
3750	($2,500,000.00)	$1,250,000.00	$5,000,000.00	$8,750,000.00	$12,500,000.00
4000	($2,000,000.00)	$2,000,000.00	$6,000,000.00	$10,000,000.00	$14,000,000.00
4250	($1,500,000.00)	$2,750,000.00	$7,000,000.00	$11,250,000.00	$15,500,000.00
4500	($1,000,000.00)	$3,500,000.00	$8,000,000.00	$12,500,000.00	$17,000,000.00
4750	($500,000.00)	$4,250,000.00	$9,000,000.00	$13,750,000.00	$18,500,000.00

DQLab

ForecastSheet

📦 **Konteks:**

Perusahaan DQFashions ingin memprediksi penjualan untuk **12 bulan ke depan** sebagai bahan perencanaan produksi dan strategi marketing.

Saat ini, data transaksi historis tersedia di file **DQLab Fashion - transaksi.xlsx**. Anda diminta mengolah data tersebut menjadi ringkasan penjualan per bulan, lalu membuat **Forecast Sheet** untuk melihat proyeksi penjualan di masa depan.

DQLab

✅ 1. Import & Rekap Data Transaksi

- Ambil data dari **DQLab Fashion - transaksi.xlsx** ke file analisis ini menggunakan **Power Query**.
- Pastikan data mencakup:
  - Tanggal Transaksi
  - Quantity (Qty)
  - Harga Produk (dari Master Produk, jika belum ada di transaksi)
- Buat kolom **Total Transaksi = Qty × Harga**

		Kode Transaksi	Urutan	Kode Produk	Jumlah Pembelian	Harga	Total Transaksi
1	7	CAB001-00001	1	ATS-0004	3	231000	693000
2	7	CAB001-00001	2	AKS-0017	3	135000	405000
3	7	CAB001-00002	2	AKS-0017	1	135000	135000
4	7	CAB001-00006	2	AKS-0017	3	135000	405000
5	7	CAB001-00002	1	ATS-0002	2	192000	384000
6	7	CAB001-00011	1	ATS-0002	2	192000	384000
7	7	CAB001-00001	3	BWH-0009	2	168000	336000

DQLab

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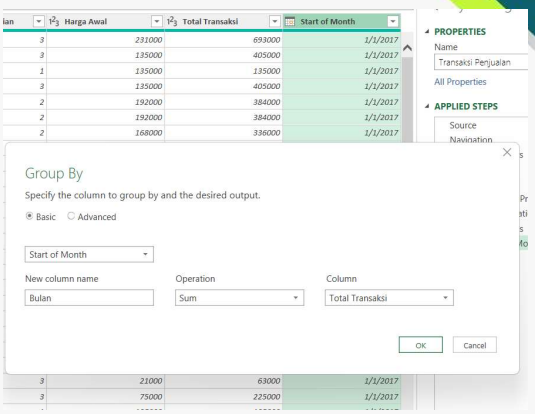
✅ 2. Buat Rekap Penjualan Bulanan

- Tambahkan kolom **Bulan (Start of Month)** dari Tanggal Transaksi
- Group By di Power Query:
  - **Group by Bulan**
  - Agregasi: Sum of Total Transaksi
- Load hasil rekapitulasi ke worksheet **ForecastSheet** ke kolom:
  - **Bulan** → Kolom **B** (B3 ke bawah)
  - **Sales** → Kolom **C** (C3 ke bawah)



	A	B	C
1			
2			
3		Bulan	Sales
4		1/1/2017	4721088000
5		2/1/2017	4082589000
6		3/1/2017	4768815000
7		4/1/2017	4692711000
8		5/1/2017	4858890000
9		6/1/2017	7943226000
10		7/1/2017	4865133000
11		8/1/2017	4943376000

	Bulan	Sales
1	1/1/2017	4721088000
2	2/1/2017	4082589000
3	3/1/2017	4768815000
4	4/1/2017	4692711000
5	5/1/2017	4858890000
6	6/1/2017	7943226000

DQLab

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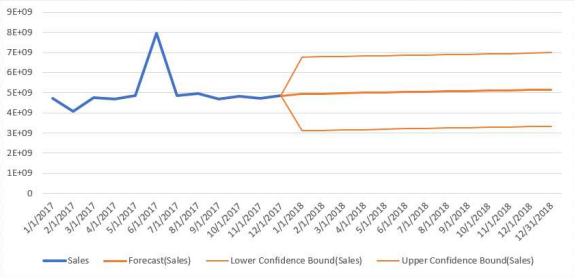
✅ 3. Forecast Penjualan dengan Forecast Sheet

1. Blok area data kolom **Bulan (B3:Bn)** dan **Sales (C3:Cn)**
2. Klik: **Data → Forecast Sheet**
3. Pilih:
   - **Line Chart**
   - Forecast End Date: +12 bulan dari data terakhir
4. Klik **Create**

Bulan	Sales	Forecast(Sales)	Lower Confidence Bound(Sales)	Upper Confidence Bound(Sales)
1/1/2017	4721088000			
2/1/2017	4082589000			
3/1/2017	4768815000			
4/1/2017	4692711000			
5/1/2017	4858890000			
6/1/2017	7943226000			
7/1/2017	4865133000			
8/1/2017	4943376000			
9/1/2017	4686954000			
10/1/2017	4824009000			
11/1/2017	4724394000			
12/1/2017	4849944000	4849944000	4.85E+09	4.85E+09
1/1/2018		4938051072	3.11E+09	6.76E+09
2/1/2018		4956740345	3.13E+09	6.78E+09
3/1/2018		4975429618	3.15E+09	6.80E+09
4/1/2018		4994118891	3.17E+09	6.82E+09
5/1/2018		5012808165	3.19E+09	6.84E+09
6/1/2018		5031497438	3.21E+09	6.86E+09
7/1/2018		5050186711	3.23E+09	6.88E+09
8/1/2018		5068875984	3.24E+09	6.89E+09
9/1/2018		5087565258	3.26E+09	6.91E+09
10/1/2018		5106254531	3.28E+09	6.93E+09
11/1/2018		5124943804	3.30E+09	6.95E+09
12/1/2018		5143633077	3.32E+09	6.97E+09
12/31/2018		5161719471	3.34E+09	6.99E+09



ForecastFunction

📝 **Latar Belakang:**

Manajemen DQFashions memerlukan analisis **forecast penjualan 12 bulan ke depan** untuk mendukung perencanaan produksi dan alokasi sumber daya.

Sebelumnya, proyeksi telah dilakukan menggunakan fitur **Forecast Sheet di Excel** yang memberikan hasil prediksi berbasis **Exponential Smoothing (ETS)**.

Namun, atasan menginstruksikan agar forecast juga dilakukan menggunakan **formula manual (fungsi FORECAST.ETS)** agar hasilnya dapat dibandingkan secara detail, terutama untuk keperluan dokumentasi metodologi forecasting yang lebih transparan.

DQLab

Bulan	Sales	Forecast(Sales)	Lower Confidenc	Upper Confidence Bound(Sales)
Jan 17	4,721,088,000			
Feb 17	4,082,589,000			
Mar 17	4,768,815,000			
Apr 17	4,692,711,000			
May 17	4,858,890,000			
Jun 17	7,943,226,000			
Jul 17	4,865,133,000			
Aug 17	4,943,376,000			
Sep 17	4,686,954,000			
Oct 17	4,824,009,000			
Nov 17	4,724,394,000			
Dec 17	4,849,944,000	4,849,944,000	4,849,944,000	4,849,944,000
Jan 18		4,938,051,072	3,113,081,690	6,763,020,453
Feb 18		4,956,740,345	3,131,762,751	6,781,717,939
Mar 18		4,975,429,618	3,150,437,425	6,800,421,811
Apr 18		4,994,118,891	3,169,103,886	6,819,133,896
May 18		5,012,808,165	3,187,760,311	6,837,856,018
Jun 18		5,031,497,438	3,206,404,875	6,856,590,000
Jul 18		5,050,186,711	3,225,035,754	6,875,337,668
Aug 18		5,068,875,984	3,243,651,125	6,894,100,844
Sep 18		5,087,565,258	3,262,249,165	6,912,881,350
Oct 18		5,106,254,531	3,280,828,052	6,931,681,010
Nov 18		5,124,943,804	3,299,385,965	6,950,501,643
Dec 18		5,143,633,077	3,317,921,085	6,969,345,070
Dec 18		5,161,719,471	3,335,834,479	6,987,604,463

Bulan	Sales
Jan 17	4,721,088,000
Feb 17	4,082,589,000
Mar 17	4,768,815,000
Apr 17	4,692,711,000
May 17	4,858,890,000
Jun 17	7,943,226,000
Jul 17	4,865,133,000
Aug 17	4,943,376,000
Sep 17	4,686,954,000
Oct 17	4,824,009,000
Nov 17	4,724,394,000
Dec 17	4,849,944,000

Master Produk | Katagori Produk | Transaksi | Scenario Summary | Scenario | GoalSeek | DataTable | Sheet6 | ForecastSheet | ForecastFunc

DQLab

Langkah-Langkah Pengerjaan Forecast Formula:

1. Menggunakan data historis penjualan bulanan (Jan 2017 – Des 2017) sebagai known data points.
2. Melakukan perhitungan forecast 12 bulan ke depan (Jan 2018 – Des 2018) dengan fungsi:
   - **FORECAST.ETS** untuk prediksi penjualan
   - **FORECAST.ETS.CONFINT** untuk menghitung confidence interval 95%
3. Hasil perhitungan dimasukkan ke dalam kolom:
   - **Forecast Sales (J)**
   - **Lower Confidence Bound (K)**
   - **Upper Confidence Bound (L)**

DQLab

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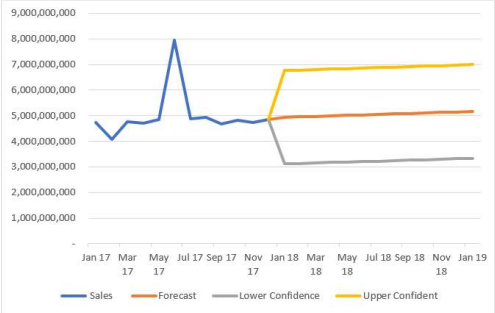
Bulan	Sales	Forecast	Lower Confidence	Upper Confident
Jan 17	4,721,088,000			
Feb 17	4,082,589,000			
Mar 17	4,768,815,000			
Apr 17	4,692,711,000			
May 17	4,858,890,000			
Jun 17	7,943,226,000			
Jul 17	4,865,133,000			
Aug 17	4,943,376,000			
Sep 17	4,686,954,000			
Oct 17	4,824,009,000			
Nov 17	4,724,394,000			
Dec 17	4,849,944,000	4,849,944,000	4,849,944,000	4,849,944,000
Jan 18		4,938,051,072	3,113,081,690	6,763,020,453
Feb 18		4,956,740,345	3,131,762,751	6,781,717,939
Mar 18		4,975,429,618	3,150,437,425	6,800,421,811
Apr 18		4,994,118,891	3,169,103,886	6,819,133,896
May 18		5,012,808,165	3,187,760,311	6,837,856,018
Jun 18		5,031,497,438	3,206,404,875	6,856,590,000
Jul 18		5,050,186,711	3,225,035,754	6,875,337,668
Aug 18		5,068,875,984	3,243,651,125	6,894,100,844
Sep 18		5,087,565,258	3,262,249,165	6,912,881,350
Oct 18		5,106,254,531	3,280,828,052	6,931,681,010
Nov 18		5,124,943,804	3,299,385,965	6,950,501,643
Dec 18		5,143,633,077	3,317,921,085	6,969,345,070
Jan 19		5,162,322,351	3,336,431,592	6,988,213,109

=FORECAST.ETS(H15, $I$3:$I$14, $H$3:$H$14)

=J15 - FORECAST.ETS.CONFINT(H15, $I$3:$I$14, $H$3:$H$14, 0.95)

=J15 + FORECAST.ETS.CONFINT(H15, $I$3:$I$14, $H$3:$H$14, 0.95)



DQLab

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Scenario

📦 **Konteks Bisnis:**

Perusahaan **DQFashions** adalah perusahaan retail yang menjual produk fashion di Indonesia. Manajemen telah menetapkan **target Revenue sebesar Rp70 Miliar** untuk tahun 2018.

Namun, untuk memastikan target laba dapat tercapai, perusahaan perlu menganalisis beberapa **skenario biaya operasional** karena adanya potensi:

1. **Kenaikan harga bahan baku (COGS)**
2. **Kenaikan biaya kantor dan administrasi karena ekspansi**
3. **Inisiatif efisiensi biaya untuk menekan pengeluaran gaji & tunjangan**

Sebagai analis keuangan, Anda diminta membantu menyusun **Forecast Laporan Laba Rugi (Profit & Loss) Tahun 2018** dengan mempertimbangkan skenario perubahan biaya.

DQLab

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📊 Asumsi Biaya terhadap Revenue (Awal):

Komponen	Persentase terhadap Revenue
COGS	30.25%
Biaya Kantor	6.05%
Biaya Admin	9.08%
Gaji/Tunjangan	30.25%
Biaya Lain	6.05%
Depreciation	14.52%

DQLab

live Class

📝 Skenario yang Diminta:

Case Name	Deskripsi
Case 1 – As-Is	Asumsi awal, tanpa perubahan biaya
Case 2 – 10% Increase	Semua komponen biaya dinaikkan sebesar 10% dari asumsi awal
Case 3 – Efficiency	Biaya Admin dan Gaji/Tunjangan dikurangi masing-masing 5%, komponen lain tetap

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Tugas Anda:

1. Inputkan **Revenue Tahun 2018 sebesar Rp70.000.000.000** di sel C4.
2. Hitung komponen biaya (COGS, Kantor, Admin, dll) di baris C7:C11 dengan rumus:
   - COGS: `=C$4 * C16`
   - Lakukan hal yang sama untuk biaya lainnya.
3. Hitung:
   - **Total Biaya (C12):** Jumlah seluruh biaya (SUM C7:C11)
   - **Net Income (C13):** Revenue – Total Biaya (`=C4 - C12`)

	A	B	C
2			
3		**Revenue**	
4		Revenue	70,000,000,000
5			
6		**COGS**	
7		**Biaya Kantor**	21,175,000,000
8		**Biaya Admin**	4,235,000,000
9		**Gaji/Tunjangan**	6,352,500,000
10		**Biaya Lain**	21,175,000,000
11		**Depreciation**	4,235,000,000
12		**Total Biaya**	**57,172,500,000**
13		**Net Income**	**12,827,500,000**
14			
15		**Asumsi**	
16		**COGS**	30.25%
17		**Biaya Kantor**	6.05%
18		**Biaya Admin**	9.08%
19		**Gaji/Tunjangan**	30.25%
20		**Biaya Lain**	6.05%
21		**Depreciation**	14.52%
22			

DQLab

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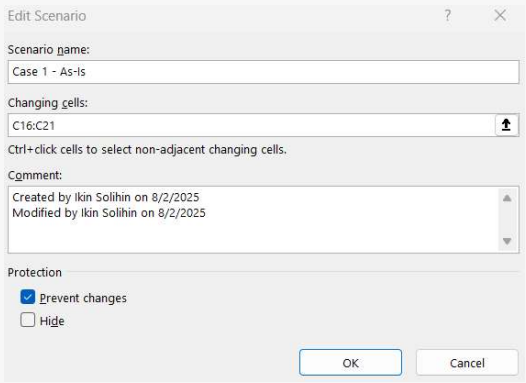
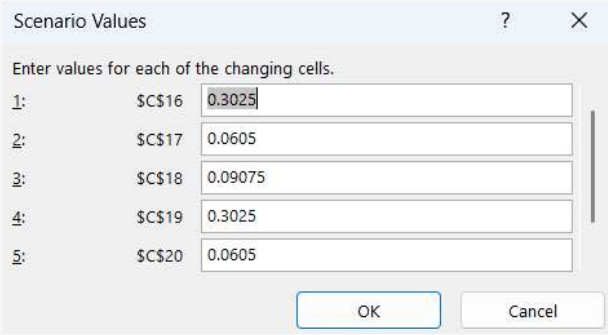
Gunakan Scenario Manager di Excel:

1. Buka **Data → What-If Analysis → Scenario Manager**
2. Tambahkan 3 skenario:
   - **Case 1 – As-Is:** Persentase biaya seperti asumsi awal (C16:C21)
   - **Case 2 – 10% Increase:** Semua biaya dinaikkan 10%
   - **Case 3 – Efficiency:** Biaya Admin & Gaji dikurangi 5%, yang lain tetap
3. Buatkan **Scenario Summary Report** untuk membandingkan:
   - Total Biaya (C12)
   - Net Income (C13)

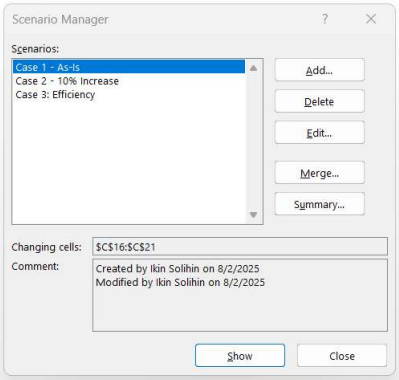




Revenue	
Revenue	70,000,000,000
COGS	
Biaya Kantor	21,175,000,000
Biaya Admin	4,235,000,000
Gaji/Tunjangan	2,852,500,000
Biaya Lain	17,675,000,000
Depreciation	4,235,000,000
Total Biaya	**50,172,500,000**
Net Income	**19,827,500,000**

Asumsi	
COGS	30.25%
Biaya Kantor	6.05%
Biaya Admin	4.08%
Gaji/Tunjangan	25.25%
Biaya Lain	6.05%
Depreciation	14.52%

case 1	As-Is
case 2	10% Increase: Semua komponen biaya dinaikan 10%
case 3	Efficiency: Biaya Admin dan Gaji dikurangi 5%, yang lain tetap



DQLab

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Revenue	
Revenue	70,000,000,000
COGS	
Biaya Kantor	21,175,000,000
Biaya Admin	4,235,000,000
Gaji/Tunjangan	2,852,500,000
Biaya Lain	17,675,000,000
Depreciation	4,235,000,000
Total Biaya	**50,172,500,000**
Net Income	**19,827,500,000**

Scenario Summary ? ×

Report type

Scenario summary

Scenario PivotTabl

Result cells:

=$C$12:$C$13

OK

Scenario Summary	Current Values:	Case 1 - As-Is	Case 2 - 10% Increase	Case 3: Efficiency
Changing Cells:				
$C$16	30.25%	30.25%	33.28%	30.25%
$C$17	6.05%	6.05%	6.66%	6.05%
$C$18	4.08%	9.08%	9.98%	4.08%
$C$19	25.25%	30.25%	33.28%	25.25%
$C$20	6.05%	6.05%	6.66%	6.05%
$C$21	14.52%	14.52%	14.52%	14.52%
Result Cells:				
$C$12	50,172,500,000	57,172,500,000	62,889,750,000	50,172,500,000
$C$13	19,827,500,000	12,827,500,000	7,110,250,000	19,827,500,000

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

DQLab

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Scenario Summary	Current Values:	Case 1 - As-Is	Case 2 - 10% Increase	Case 3: Efficiency
Changing Cells:				
COGS	30.25%	30.25%	33.28%	30.25%
Biaya Kantor	6.05%	6.05%	6.66%	6.05%
Biaya Admin	4.08%	9.08%	9.98%	4.08%
Gaji/Tunjangan	25.25%	30.25%	33.28%	25.25%
Biaya Lain	6.05%	6.05%	6.66%	6.05%
Depreciation	14.52%	14.52%	14.52%	14.52%
Result Cells:				
Total Biaya	50,172,500,000	57,172,500,000	62,889,750,000	50,172,500,000
Net Income	19,827,500,000	12,827,500,000	7,110,250,000	19,827,500,000

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

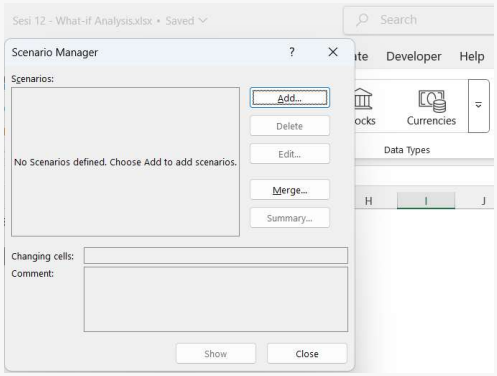
DQLab

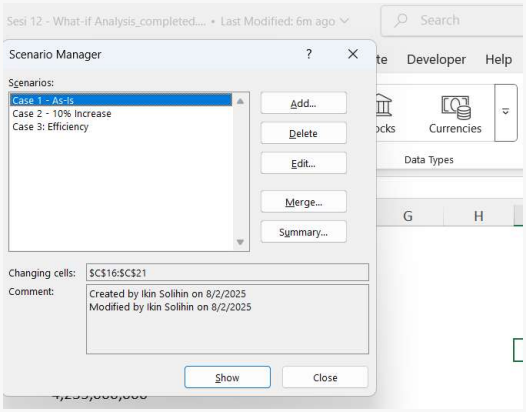
live class

Merge

Sesi 12 - What-if Analysis.xlsx

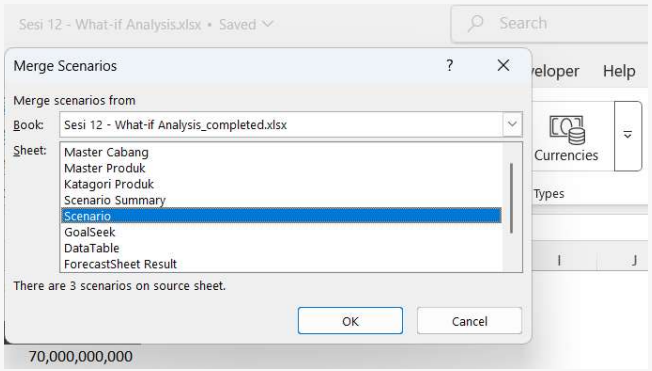
Sesi 12 - What-if Analysis_completed.xlsx

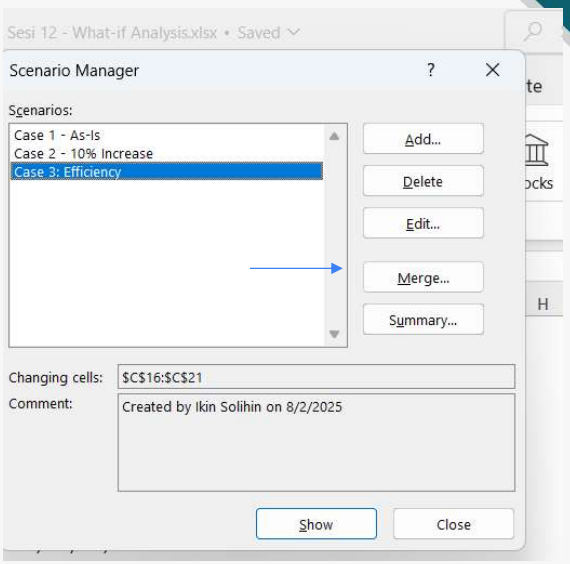


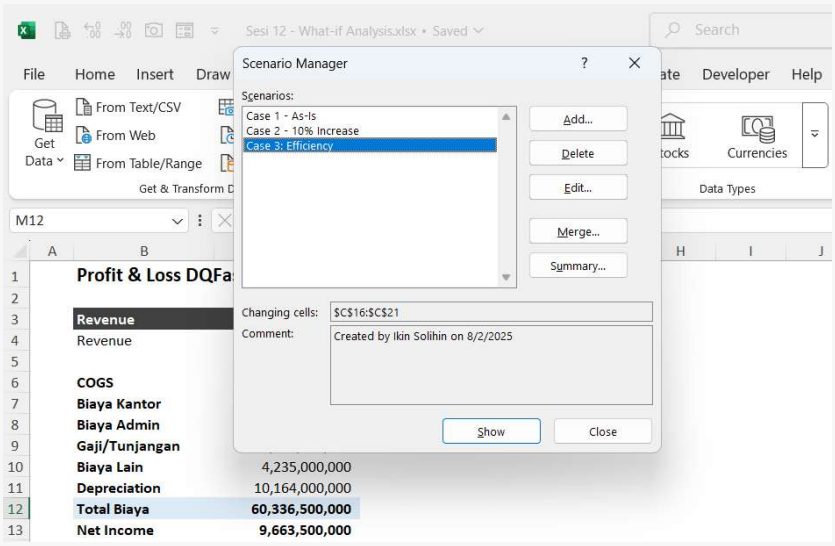


DQLab

live class







All Scenarios are copied to the file

Pengenalan Misi

Misi dapat diakses di: https://academy.dqlab.id/